Can we detect CO?

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1. Can we detect CO?

This presentation answers two questions: can we detect CO through the channel, and why is its absorption a smooth curve rather than a set of sharp bands? We begin with the physical mechanism, then connect it to the signal we would have to measure.

2. Can we detect it? The answer depends on what is known

Yes: in the ideal case, CO changes the channel and a perfectly matched reference lets us recover its added absorption. But if the atmosphere is completely unknown, the same observation can have several explanations. We have not yet established a practical detection limit.

3. The electric field can excite CO rotation

CO has a permanent electric dipole, so the wave's electric field can drive changes in its rotational state. The energy difference sets the resonance frequency, and the allowed transitions give the sequence of line centers. This explains where the peaks occur, but not yet their width.

4. Collisions interrupt the molecular phase

Collisions perturb the molecular environment and interrupt the phase of its response to the field. A response that loses coherence quickly cannot have an infinitely sharp frequency. In the simple Lorentz model, the first CO line at one atmosphere corresponds to an effective coherence time of about 66 picoseconds.

5. Thermal motion gives different Doppler shifts

Even at one temperature, molecules have different velocities along the beam, so they see slightly different frequencies through the Doppler effect. The thermal velocity distribution produces a Gaussian profile. For this CO transition at 296 kelvin, its full width is about 0.268 megahertz.

6. At atmospheric pressure, collisions set most of the width

The Voigt profile combines Lorentz broadening from collisions with Gaussian broadening from thermal motion. Here, pressure broadening is about eighteen thousand times larger than the Doppler width, so collisions dominate. Increasing pressure spreads the same normalized line area over a wider frequency range.

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7. Why the curve in your image is smooth

The curve in the original image was calculated for acetonitrile, not CO. It adds broadened line contributions through an atmosphere whose pressure and temperature change with height, so the result can look smooth across a small frequency window. That smooth appearance does not by itself prove continuum absorption or identify the gas.

8. Three physical steps connect a molecule to the channel

We first add the individual molecular lines to obtain the absorption cross section. We then multiply by molecular density and integrate along the path to obtain optical depth. Beer-Lambert converts optical depth into transmission: the exponent is minus tau for power and minus tau over two for channel amplitude.

9. CO provides a sequence of spectral features

The CO model produces eight main rotational features between 60 and 1000 gigahertz. Their heights differ because transition strengths, level populations and linewidths differ. We should compare the whole spectral pattern with the background, rather than assume that the tallest peak is automatically the best sensing band.

10. Observe the chemical loss through a channel ratio

We measure the channel at each frequency and path length, then compare it with a matched reference without the added CO. Dividing their magnitudes removes the shared channel factors, and the logarithm gives the extra CO optical depth. If the reference contains ambient CO, we retrieve the enhancement above that reference.

Key equation: $\tau_c = -2 \ln(|h| / |h_0|)$

11. Path length changes the gas column that the wave crosses

Changing the path length tells us how much of the gas cloud the wave has crossed. Here the extra CO loss increases only between 20 and 40 metres, then becomes constant. The integrated signal gives a gas column; concentration also requires the cloud length or additional spatial information.

12. The predicted signal is small, even when it is identifiable

At 115 gigahertz, the amplitude change in this example is only about 0.33 parts per million. It grows at higher CO lines, but some of those frequencies also suffer much greater atmospheric loss. These numbers tell us the signal scale an instrument would have to resolve; they do not demonstrate that an instrument can resolve it.

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13. A small atmospheric mismatch can dominate the CO loss

At the 345.8 gigahertz CO line, a one percent increase in water partial pressure over the full path changes the loss about 235 times more than our CO cloud does. This is a chosen sensitivity example, not a measured humidity error. It shows why we must model the background, while different spectral shapes may still allow the two effects to be separated.

14. CO must be distinguishable from atmospheric changes

With a completely free atmospheric spectrum, there are many ways to divide the total loss between air and CO. If we constrain the atmosphere to known physical parameters, we can instead compare their effects across frequency. CO is distinguishable only when its response cannot be reproduced by the allowed background changes.

15. What we can conclude now

Our answer is yes under the stated ideal assumptions: the added CO absorption is observable through the reference ratio. Its smooth spectral shape is caused by finite molecular coherence and thermal motion around discrete transitions. The next step is to quantify reference accuracy and test whether CO remains distinguishable before making a practical detection claim.

16. Appendix: the law behind the smooth line

The continuous profile follows directly from the frequency response of a decaying molecular coherence. An exponential decay produces a Lorentzian, whose width grows as the coherence time becomes shorter. Adding the distribution of Doppler shifts gives the Voigt profile, without introducing receiver noise.

17. Sources and numerical evidence

The molecular explanation is supported by NIST and MIT spectroscopy references. The numbers use HITRAN line parameters, JPL line positions and an ITU atmospheric model, with the assumptions recorded in the project. They are model predictions, not field measurements.

Slides 16 and 17 are supporting material for questions. Scientific qualifications and links are in the presentation notes and the research record.